Human-Like Interactive Lane-Change Modeling Based on Reward-Guided Diffusive Predictor and Planner

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***Abstract*— Lane changing presents a dynamic scenario characterized by intricate interactions among vehicles. Within mixed-autonomy traffic environment, modeling a human-like lane-change trajectory enables human drivers to better understand and predict autonomous vehicles' behaviors, thereby enhancing road safety and travel efficiency. In this study, we achieve human-like interactive lane-change modeling based on a novel framework named *Diff-LC*. The human-like modeling of LCV behaviors relies on an advanced diffusive planner, and the implemented trajectory is selected based on the recovered LCV reward function learned through Multi-Agent Adversarial Inverse Reinforcement Learning (MA-AIRL). To account for interactions between FVs and LCVs, we further employ a diffusive predictor to forecast future behaviors of FVs conditioned on both historical and planned trajectories. Additionally, we leverage the recovered reward function of FVs to enable controllable prediction of trajectories. In the experimental part, we begin by analyzing the significance of features in the recovered reward functions and then proceed to compare the distinctions between the LCV and the FV. To validate the effectiveness of the proposed framework, we compare the diffusive predictor and planner with several state-of-the-art methods. The results demonstrate that motions planned by Diff-LC closely reach the intended positions with small displacement errors and exhibit highly similar speed and jerk distributions to those of human drivers. We also conduct a dynamic simulation to evaluate Diff-LC's performance across different traffic conditions. Finally, we explore customized generation using the Diffusion Posterior Sampling method. The codes can be found at https://github.com/zeonchen/Diff-LC/.**

***Index Terms*— Lane changing, human-like modeling, diffusion model, trajectory prediction.**

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I. Introduction

As a fundamental driving behavior, drivers implement lane-changing (LC) maneuvers to transition from one lane to another. Normally, LC maneuvers can be further divided into two types: mandatory and discretionary LC. Mandatory LC happens in merging scenarios or when drivers adhere to pre-determined routes, while drivers implement discretionary LC to improve travel efficiency. Compared to mandatory LC, discretionary LC is more challenging to depict due to its stochasticity [1], [2]. As Fig. 1 shows, a typical LC scenario involves four vehicles: a lane-changing vehicle (LCV) and its leading vehicle (LV_{cur}), a following vehicle (FV) in the target lane and its leading vehicle (LV_{tar}). When performing LC maneuvers, the LCV should take into account the constraints of both LVs and engage in active interaction with the FV.

However, improper LC behaviors lead to traffic congestion or even crashes [3], [4]. Hence, it is essential to accurately model and depict LC maneuvers to alleviate congestion as well as reduce traffic crashes. Additionally, with the thriving of autonomous driving, it is inevitable to have mixed scenarios with the co-existence of both human-driven vehicles (HV) and autonomous vehicles (AV) before fully automated mobility [5]. By mimicking human behaviors for AV, HV will find it easier to interact and understand the AV. Furthermore, human-like modeling facilitates the development of realistic traffic simulators [6].

To recover human-like driving strategies from naturalistic datasets, previous studies have endeavored to employ behavior cloning [7] and inverse reinforcement learning (IRL) [8]. Behavior cloning directly learns from human experts in a supervised learning manner, but it is susceptible to dataset bias, overfitting, and generalization issues [9]. Compared to behavior cloning, IRL initially recovers the reward function from human demonstrations and subsequently learns the driving policy based on RL algorithms. Consequently, IRL can derive a more authentic driving policy.

Nevertheless, current studies are subject to several limitations. First, while IRL can successfully recover a high-quality reward function, the learned policy may still produce unrealistic behaviors, potentially violating the human likeness. Second, current studies employing IRL/RL failed to accurately capture interactions among vehicles. Some approaches merely

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introduced hand-crafted interactive terms into the reward function, which may not fully represent real-world interactions. Last, most prior studies have primarily focused on the behavior of the LCV while largely neglecting the behavior of the FV, despite the significant influence of the cut-in maneuver on the FV's actions.

To fill the above gaps, we propose a novel framework named ***Diff-LC*** to achieve human-like discretionary lane-changing modeling. To address the second limitation, we begin the training phase with Multi-Agent Adversarial Inverse Reinforcement Learning (MA-AIRL) to recover reward functions for both the LCV and FV, enabling the model to implicitly learn interactive reward functions from the data. Leveraging the recent surge in popularity and remarkable performance of diffusion models in various fields [10], [11], we harness this novel and powerful technique to develop a planner and predictor. To tackle the first limitation, we use a diffusive planner to achieve human-like trajectory planning during lane changes. Additionally, a diffusive predictor is employed to anticipate the potential behavior of the FV based on the planned actions of the LCV, addressing the third limitation. These diffusive models are trained using naturalistic datasets. Subsequently, we train a goal predictor and a reward guide to enhance the overall inference process. During the testing phase, we sample various goal states around the predicted goal. Subsequently, we generate planned trajectories for the LCV for each goal state and predict corresponding trajectories for the FV with the guidance of the reward guide. Finally, we evaluate these trajectories using the reward function of the LCV. In contrast to prior research, the planned trajectories are crafted by a data-driven diffusive planner and assessed using the recovered reward function, diverging from the conventional learning approach of RL agents. Furthermore, the interaction between the FV and the LCV is encapsulated through MA-AIRL, and the evaluation stage comprehensively incorporates the behavior of both vehicles.

The main contributions of our study are as follows:

- We employ MA-AIRL to recover reward functions for both LCV and FV from human demonstrations, thereby implicitly capturing interactions among vehicles without relying on hand-crafted interaction terms.
- We develop a data-driven planner and predictor, based on modern diffusion models, to achieve human-like trajectory planning that accounts for interactions between LCVs and FVs. The generation process is guided by the interactive reward function derived from MA-AIRL.
- We conduct extensive experiments to verify the effectiveness of our framework. The results demonstrate the superiority of the Diff-LC framework, not only in reaching goal states but also in achieving human-like speed and jerk distributions.

The remainder of the paper is organized as follows. We briefly introduce related works in Section II. Afterwards, the proposed method is revealed in Section III. Section IV thoroughly introduces experiments for evaluation, results and discussions. At last, we summarize the overall paper in Section V.

II. Related Work

In this section, we briefly introduce topics related to lane-changing modeling and human-like driving behavior modeling.

A. Decision-Making in Lane-Changing Scenarios

LC scenarios entail intricate longitudinal and lateral movements, as well as interactions among vehicles. Recent studies have leaned towards solving decision-making challenges through data-driven methods, particularly leveraging RL techniques.

For instance, [12] designed the reward function by combining individual efficiency with overall efficiency for harmony. Then, a deep RL method was applied to make high-level LC decisions followed by low-level controllers. [13] used reinforcement learning to decide LC time, followed by a unified trajectory planning model. The reward function was relatively simple and mainly focused on travel speed. To handle merging scenarios in mixed-traffic highways, [14] used MARL to coordinate connected AVs with shared parameters and local rewards. Similarly, [15] also utilized MARL to achieve collaborative decision-making among vehicles. The MARL was used for upper-level decision-making, while a right-of-way assignment was applied by the authors to ensure the safety of LC behaviors. Reference [16] employed a MARL method to coordinate behaviors of connected AVs based on field experiments, and the proposed methods effectively alleviated congestion in the mandatory LC scenario.

Nevertheless, while driving policies obtained from the aforementioned frameworks achieved high travel efficiency and collision avoidance, they may not necessarily guarantee human-like behavior.

B. Human-Like Driving Behavior Modeling

Previous studies employed two paradigms to achieve human-like driving behavior modeling. The first paradigm develops models based on traditional planning algorithms or kinematic models. For instance, [17] designed a cost function based on human preferences for the evaluation of trajectory candidates. Reference [18] developed a data-driven planner using a general regression neural network to derive human-like trajectories. A personalized human-like lane-changing trajectory planning method was proposed in [19] by considering both the traffic environment and drivers' psychological characteristics. Reference [20] simultaneously considered drivers' visual-preview steering characteristics, risk perception and the degree of aggressiveness to achieve human-like planning. Reference [21] fitted the bicycle model into the human driving data and investigated the influence of sampling rates. Reference [22] proposed a preview-based human-like trajectory planning model by considering drivers' visual attention mechanism. However, these methods are limited to the accuracy and expressiveness of base algorithms or models.

The second paradigm advances model development through imitation learning, employing data-driven approaches such as behavior cloning and inverse reinforcement learning that

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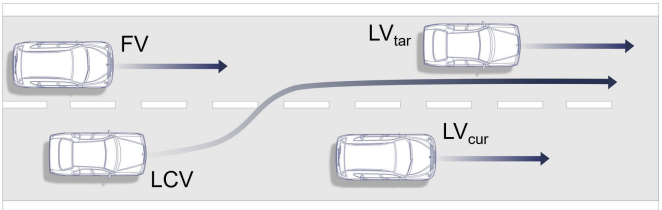

Fig. 1. A typical lane-changing scenario. There are a lane-changing vehicle and a following vehicle, while two leading vehicles are in the current and target lanes.

foster the creation of more authentic models. [23] applied Convolutional Neural Networks (CNN) to control steering angles based on images. Reference [24] combined CNN and Variational Auto-Encoder (VAE) to achieve end-to-end control of autonomous vehicles. Reference [25] employed a Generative Adversarial Network (GAN) to generate semantic data, which enhances knowledge transfer to new weather conditions during driving. To handle domain shift scenarios, a recent study [26] proposed an off-policy behavior cloning method using task knowledge distillation.

For IRL methods, [27] used an RL planner with a learned reward function induced by IRL to achieve human-like autonomous driving. To enhance sampling efficiency and incorporate trajectory uncertainties, [28] proposed an efficient sampling-based maximum entropy IRL to learn human preferences. Additionally, [29] employed maximum entropy IRL to recover reward functions from naturalistic driving data. Afterwards, the trajectory was selected from various candidates based on the learned reward function. To model interactions among vehicles, the authors considered incorporating the decelerations of the environment vehicles influenced by the ego vehicle in the reward function. Reference [30] proposed a two-step method to build customized automated LC systems. IRL was used to learn the reward functions of each driving style. Similarly, [31] derived reward functions of each intention during LC maneuvers. The predicted trajectories were selected from various candidates.

Reference [32] assumed the AVs' future trajectories were conditioned on the future behaviors of surrounding vehicles and planned trajectories. The IRL was used to select the most suitable trajectory. Reference [33] proposed an interaction-aware planning method based on deep IRL. The authors employed a reward network to extract human preferences from demonstrations. To encompass the interaction between the ego vehicles and the following vehicles, a feature-based IRL was used to simulate the trajectories of the following vehicles influenced by the ego vehicles. Reference [34] applied IRL to analyze the car-following behaviors when HVs follow AVs. Compared to LC maneuvers, car-following is a reaction process and does not involve interactions. Reference [35] utilized parameter-sharing Generative Adversarial Imitation Learning (GAIL) to concurrently simulate multiple vehicles. In contrast to single-agent GAIL, the multi-agent variant adeptly reproduces the driving behaviors of human drivers, even with a number of controlled vehicles. Recently, [36] modified Adversarial IRL (AIRL) with goal-conditioned and confidence-aware mechanism. The method was applied in the merging scenario and achieved better performance than previous imitation learning approaches.

TABLE I
ABBREVIATIONS USED IN THIS PAPER

Abbreviation	Full Name
LC	Lane changing
LCV	Lane-changing vehicle
LV$_{cur}$	Leading vehicle on the current lane
LV$_{tar}$	Leading vehicle on the target lane
FV	Following vehicle
HV	Human-driven vehicle
AV	Autonomous vehicle
IRL	Inverse reinforcement learning
RL	Reinforcement learning
MA-AIRL	Multi-agent adversarial IRL
LSBRE	Logistic stochastic best response equilibrium
DDPM	Denoising diffusion probabilistic model
DDIM	Deep diffusion implicit model
GRU	Gated Recurrent Unit
PPO	Proximal policy optimization
MLP	Multi-layer perceptron
ADE	Average displacement error
FDE	Final displacement error
IDM	Intelligent driver model
DPS	Diffusion posterior sampling

III. METHODOLOGY

A. Problem Definition

Table I presents the abbreviation used in this paper. The goal of this study is to achieve human-like modeling of autonomous vehicles (AVs) based on collected human demonstrations. As Fig. 1 shows, the LC scenario is highly interactive and involves the dynamic interplay between LCV and FV. Given the current vehicle state s_0^* and potential goal states $s_{T,j}^*$, the LCV first plans M trajectories $\{\mathbf{p}_j\}_{j=1}^M$ and $\mathbf{p}_j \sim p(\mathbf{p}_j|s_0^*, s_{T,j}^*)$, and future trajectories of surrounding vehicles can be predicted conditioned on both historical and planned trajectories. Indeed, the movements of FVs are influenced not only by LCVs but also by other surrounding vehicles, such as LVs. However, in our emphasis on the interaction process between LCVs and FVs during LC scenarios, we make the assumption that the future trajectories of LVs can be accurately predicted, as they primarily engage in typical car-following maneuvers. We denote the predicted distribution as $p(\mathbf{f}_j|\mathbf{h}_{FV}, \mathbf{h}_{LCV}, \mathbf{h}_{LV_{cur}}, \mathbf{h}_{LV_{tar}}, \mathbf{p}_j)$, where $\mathbf{f}_{(\cdot)}$ and $\mathbf{h}_{(\cdot)}$ are the predicted and historical trajectories of the FV and four vehicles. Once M pairs of predicted and planned trajectories $\{\mathbf{p}_j, \mathbf{f}_j\}_{j=1}^M$ are obtained, a human-like reward function is utilized to select the optimal planned trajectory for the LCV.

B. Framework Overview

As Fig. 2 shows, our Diff-LC has model training and inference phases. During the training phase, we initially utilize MA-AIRL to derive the reward functions for both the LCV and the FV. Next, we employ human demonstrations to train both the diffusive planner and predictor. To enhance sampling efficiency, we further train a goal predictor. Subsequently, a reward guide is trained utilizing the FV reward function and a noisy module.

In the inference phase, we begin by sampling M goal states based on the predicted goal and generating human-like planned

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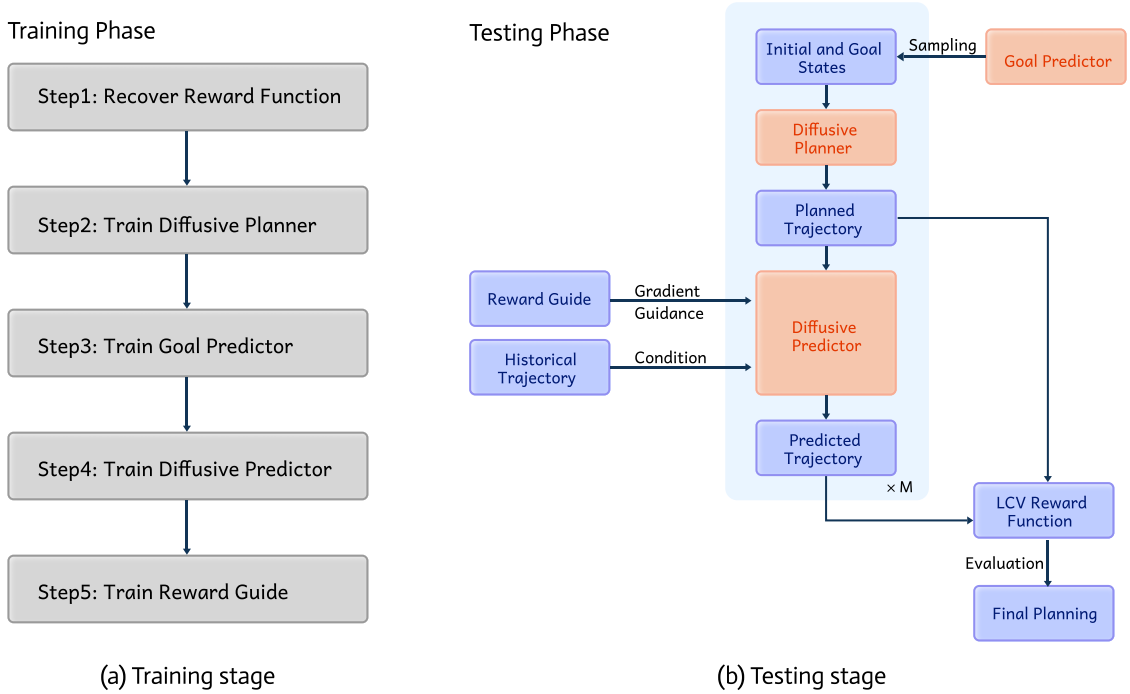

Fig. 2. Overview of the Diff-LC framework including training and testing phases.

trajectories for the LCV. Subsequently, the future trajectories of the FV are predicted conditioning on historical trajectories and planned trajectories. Additionally, we utilize the reward guide to attain human-like predictions. Finally, the M pairs of trajectories are evaluated by the LCV reward function, and the best one is selected for implementation. In the following sections, we will introduce each component in detail.

C. Reward Recovery Based on MA-AIRL

The interaction between the LCV and the FV can be deemed as a Markov game [37]. A Markov game is defined as $(\mathcal{S}, \{\mathcal{A}_i\}_{i=1}^N, P, \eta, \boldsymbol{r})$, where $\mathcal{S}$ is the set of states, N is the number of agents, $\mathcal{A}_i$ is the action set of agent i, P is the transition process, η is the distribution of initial states, and $\boldsymbol{r}$ denotes all reward functions of agents. Given the collected driving trajectories $\mathcal{D}$ of human drivers, we would like to recover the reward functions of both LCV and FV. Afterwards, the recovered reward functions can be used for trajectory evaluation.

Specifically, the reward function for either LCV or FV is denoted as $r_{(\cdot)} : \mathcal{A}_{LCV} \times \mathcal{A}_{FV} \rightarrow \mathbb{R}$, and the MA-AIRL aims to find the logistic stochastic best response equilibrium (LSBRE) in an adversarial manner [38]. For the discriminator parameterized D by ω, we can train the model with:

$$L_{disc} = \max_{\boldsymbol{\omega}} \mathbb{E}_{\boldsymbol{\pi}_E}\left[\sum_{i=1}^{N} \log \frac{\exp\left(f_{\omega_i}(s,a)\right)}{\exp\left(f_{\omega_i}(s,a)\right) + q_{\theta_i}(a_i \mid s)}\right] + \mathbb{E}_{\boldsymbol{q}_{\boldsymbol{\theta}}}\left[\sum_{i=1}^{N} \log \frac{q_{\theta_i}(a_i \mid s)}{\exp\left(f_{\omega_i}(s,a)\right) + q_{\theta_i}(a_i \mid s)}\right] \quad (1)$$

where $\boldsymbol{\pi}_E$ is the expert policy; f_{ω_i} is the discriminator of agent i; q_{θ_i} is the approximated expert policy.

The generators parameterized by θ can be trained with:

$$L_{gen} = \max_{\theta} \mathbb{E}_{\boldsymbol{q}_{\boldsymbol{\theta}}}\left[\sum_{i=1}^{N} \log\left(D_{\omega_i}(s,a)\right) - \log\left(1 - D_{\omega_i}(s,a)\right)\right] = \mathbb{E}_{\boldsymbol{q}_{\boldsymbol{\theta}}}\left[\sum_{i=1}^{N} f_{\omega_i}(s,a) - \log\left(q_{\theta_i}(a_i \mid s)\right)\right] \quad (2)$$

After training, we can derive the estimated reward function g_{ω_i} as:

$$g_{\omega_i}(s_t, a_t) = f_{\omega_i,\phi_i}(s_t, a_t, s_{t+1}) - \gamma h_{\phi_i}(s_{t+1}) + h_{\phi_i}(s_t) \quad (3)$$

where γ is the discount factor, g_{ω_i} is the estimated reward function, and h_{ϕ_i} is the potential function. The authors of MA-AIRL also discovered that eliminating actions can further enhance learning performance. Hence, we only take s_t as the input of g_{ω_i} and f_{ω_i,ϕ_i}.

In this study, we define the Markov game between LCV and FV as follows:

- State $\mathcal{S}$: the states of both LCV and FV include positions of LCV and the differences of positions and velocities between LCV and the other three vehicles, i.e. $s_t \in \mathbb{R}^{16}$. Therefore, a trajectory with length T can be defined as $\tau = \{s_0, \ldots, s_T\}$.
- Action $\mathcal{A}_{LCV}$ and $\mathcal{A}_{FV}$: acceleration of both vehicles in two directions, i.e. $a^{\mathrm{LCV}} = (a_x^{\mathrm{LCV}}, a_y^{\mathrm{LCV}})$, $a^{\mathrm{FV}} = (a_x^{\mathrm{FV}}, a_y^{\mathrm{FV}})$.
- Transition P: we use the following equations to update the states of LCV and FV, and the state transition of LVs are based on authentic data:

$$v_{t+1,x}^{(\cdot)} = v_{t,x}^{(\cdot)} + a_{t,x}^{(\cdot)} \cdot \Delta t \quad (4)$$

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$$v_{t+1,y}^{(\cdot)} = v_{t,y}^{(\cdot)} + a_{t,y}^{(\cdot)} \cdot \Delta t \tag{5}$$

$$l_{t+1,x}^{(\cdot)} = l_{t,x}^{(\cdot)} + v_{t,x}^{(\cdot)} \cdot \Delta t \tag{6}$$

$$l_{t+1,y}^{(\cdot)} = l_{t,x}^{(\cdot)} + v_{t,y}^{(\cdot)} \cdot \Delta t \tag{7}$$

where Δt is the update time.

Different from [12] and [33] that defined interactions manually, our method learns interactive reward functions from human drivers. Afterwards, we will use the learned reward functions for guidance and evaluation.

D. Diffusive Planner and Predictor

1) Denoising Diffusion Model: As a powerful generative model, the main concept of the diffusion model is learning to reverse the destruction process for sample generation [39]. In the study, we develop our diffusive planner and predictor based on Denoising Diffusion Probabilistic Model (DDPM) [40]. In detail, DDPM consists of two stages. In the first stage, we manually define a Markov chain to add noise step by step, which is also called *forward process*. Specifically, for a sample $\tau_0 \sim q(\tau_0)$, we gradually add Gaussian noise based on pre-scheduled scheme:

$$q(\tau_{1:K}|\tau_0) = \prod_{k=1}^{K} q(\tau_k|\tau_{k-1}) \tag{8}$$

$$q(\tau_k|\tau_{k-1}) = \mathcal{N}(\tau_k; \sqrt{1-\beta_t}\tau_{k-1}, \beta_k \mathbf{I}) \tag{9}$$

Denote $\alpha_k = 1 - \beta_k$ and $\bar{\alpha}_k = \prod_{k=1}^{K} \alpha_k$, we can derive:

$$q(\tau_k|\tau_0) = \mathcal{N}(\tau_k; \sqrt{\bar{\alpha}_k}\tau_0, (1-\bar{\alpha}_k)\mathbf{I}) \tag{10}$$

In the second stage, we aim to recover the original sample τ_0 from noises sampled from a standard Gaussian distribution, denoted as $p(\tau_K) = \mathcal{N}(\tau_K; 0, \mathbf{I})$. This stage is also named *reverse process*. Different from the forward process, we utilize a learnable model with parameter ξ to generate mean and variance:

$$p_\xi(\tau_{0:K}) = p(\tau_K) \prod_{k=1}^{K} p_\xi(\tau_{k-1}|\tau_k) \tag{11}$$

$$p_\xi(\tau_{k-1}|\tau_k) = \mathcal{N}(\tau_{k-1}; \mu_\xi(\tau_k, k), \Sigma_\xi(\tau_k, k)) \tag{12}$$

In DDPM, we train the model by optimizing the variational lower bound:

$$-\log p_\xi(\tau_0) \leq -\log p_\xi(\tau_0|\tau_1) + \mathrm{KL}(q(\tau_K|\tau_0)||p(\tau_K)) + \sum_{k=2}^{K} \mathrm{KL}(q(\tau_{k-1}|\tau_k, \tau_0)||p_\xi(\tau_{k-1}|\tau_k)) \tag{13}$$

The authors of DDPM further simplified the objective function as:

$$L_{diff} = \mathbb{E}_{\epsilon,\tau_0,k}||\epsilon - \epsilon_\xi(\sqrt{\bar{\alpha}_t}\tau_0 + \sqrt{1-\bar{\alpha}_t}\epsilon, k)||^2 \tag{14}$$

where $\epsilon \sim \mathcal{N}(0, \mathbf{I})$.

The diffusive planner and predictor in our study operate on different states. Despite a slight abuse of notation, we maintain consistency in notation throughout the following sections, while explicitly specifying the differences in each section.

2) Diffusive Planner: In our diffusive planner, the planning process of a vehicle aims to find a sequence of actions $a_{0:T}^*$ that maximizes the overall rewards:

$$a_{0:T}^* = \mathrm{argmax}_{a_{0:T}} \sum_{t=0}^{T} r(s_t, a_t) \tag{15}$$

where T is the planning horizon, and $a_{(\cdot)}$ indicates accelerations of vehicles.

Inspired by [41], the diffusive planner generates planned trajectories based on initial and goal states (denoted as s_0^* and s_T^*), framing the task as an inpainting problem. Please note that here the state merely includes the speed, location and acceleration of the LCV. Specifically, when we sample from the unperturbed reverse process, the sampled values should be replaced with s_0^* and s_T^*. Given that the local receptive field is instrumental in enforcing local consistency throughout a single denoising step, we employ a U-Net architecture featuring residual blocks as the backbone model for the diffusive planner:

$$\tau_k = \mathrm{Mish}\big(\mathrm{GN}(\mathrm{Conv1d}(\tau_k))\big) \tag{16}$$

where $\mathrm{Mish}(\cdot)$ is the Mish nonlinearities; $\mathrm{GN}(\cdot)$ is the group normalization; and $\mathrm{Conv1d}(\cdot)$ is the 1-d convolutional layer. Fig. 3 (a) shows the model structure of our diffusive planner. After training, we can simultaneously generate M planned trajectories (denoted as $\mathbf{p}$) by sampling various goal conditions. Algorithm 1 presents the training process of the diffusive planner.

Algorithm 1 Diffusive Planner Training Process

while *not done* **do**
  $\tau_0 \sim q(\tau_0)$
  $k \sim \mathrm{Uniform}(\{1, \ldots, K\})$
  $\epsilon \sim \mathcal{N}(0, \mathbf{I})$
  $\tau_k \sim \mathcal{N}(\tau_k; \sqrt{\bar{\alpha}_k}\tau_0, (1-\bar{\alpha}_k)\mathbf{I})$
  $s_{k,0}, s_{k,T} \leftarrow s_{0,0}, s_{0,T}$
  Update model parameters based on gradient:
  $\nabla_\xi \left\| \boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\xi \left(\sqrt{\bar{\alpha}_t}\tau_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}, k\right) \right\|^2$
end while

3) Goal Predictor: The diffusive planner expects an accurate goal state s_T rather than simply sampling from a pre-determined range. To enhance the sampling efficiency, we predict the goal state of the LCV with a GRU-based network based on the historical trajectories of four involved vehicles:

$$\hat{s}_T = \mathrm{MLP}\big(\mathrm{GRU}(\mathbf{h}_{LCV} \oplus \mathbf{h}_{FV} \oplus \mathbf{h}_{LV_{cur}} \oplus \mathbf{h}_{LV_{tar}})\big) \tag{17}$$

where $\mathrm{MLP}(\cdot)$ indicates a feed forward neural network; $\mathrm{GRU}(\cdot)$ is the GRU network; $\oplus$ indicates concatenating operation.

4) Diffusive Predictor: Once the planned trajectories of the LCV are obtained, it becomes feasible to predict the future movements of the FV, thereby transforming the DDPM into a conditional diffusion process. Fig. 3 (b) shows the model structure of our diffusive predictor. Here the state refers to the speed, location, and acceleration of the FV.

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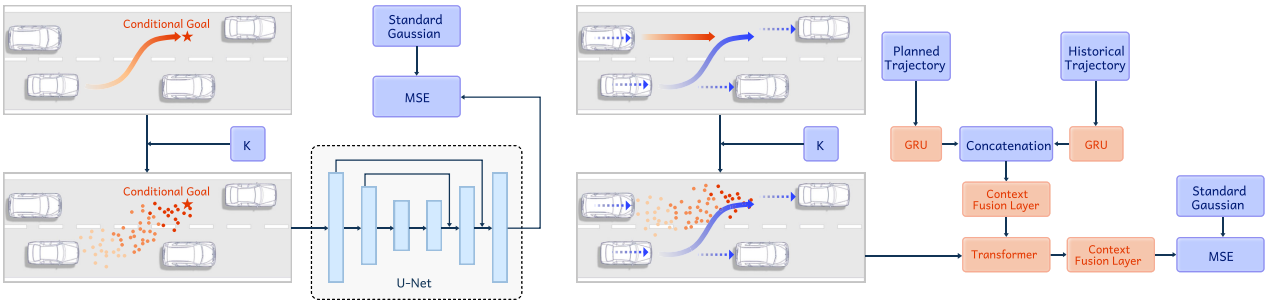

(a) Model structure of diffusive planner (b) Model structure of diffusive predictor

Fig. 3. Model structure of diffusive planner and predictor.

For the condition model, we first encode the historical and planned trajectories into embeddings via GRU:

$$\mathbf{e}_p = \mathrm{GRU}(\mathbf{p}) \tag{18}$$

$$\mathbf{e}_h = \mathrm{GRU}(\mathbf{h}_{LCV} \oplus \mathbf{h}_{FV} \oplus \mathbf{h}_{LV_{cur}} \oplus \mathbf{h}_{LV_{tar}}) \tag{19}$$

Both embeddings $\mathbf{e}_p$ and $\mathbf{e}_h$ are regarded as conditions. Then, we fuse the conditions with the noisy trajectories in the context fusion layers via a gate mechanism:

$$\mathbf{c} = \mathbf{e}_p \oplus \mathbf{e}_h \oplus \mathrm{PE}(k) \tag{20}$$

$$\tau_k = \mathbf{W}^p \tau_k \odot \sigma(\mathbf{W}^c \mathbf{c}) + \mathbf{W}^b \mathbf{c} \tag{21}$$

where $\mathrm{PE}(\cdot)$ is the position embedding layer as introduced in [42], transforming the diffusion step k into an embedding; $\mathbf{W}^p$, $\mathbf{W}^c$, and $\mathbf{W}^b$ are learnable parameters; $\odot$ indicates element-wise multiplication.

Afterwards, we input the trajectories into a Transformer encoder with time encoding [43]:

$$\tau_k = \mathrm{Transformer}(\tau_k + \mathrm{TE}(\tau_k)) \tag{22}$$

where $\mathrm{TE}(\cdot)$ is the temporal encoding layer, introducing time information in the trajectory.

Last, we re-implement the context fusion layers to further enhance the conditional information similar to Eq. 21. During training, the authentic trajectories from datasets are regarded as the planned trajectories. Algorithm 2 shows the training process of the diffusive predictor.

Algorithm 2 Diffusive Predictor Training Process

while *not done* **do**
  $\tau_0 \sim q(\tau_0)$
  $k \sim \mathrm{Uniform}(\{1, \ldots, K\})$
  $\epsilon \sim \mathcal{N}(0, \mathbf{I})$
  Update model parameters based on gradient:
    $\nabla_\xi \left\| \boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\xi \left(\sqrt{\bar{\alpha}_t} \tau_0 + \sqrt{1 - \bar{\alpha}_t} \boldsymbol{\epsilon}, k \right) \right\|^2$
end while

5) Reward Guide Training: The recovered reward function $r_{\mathrm{FV}}(\cdot)$ of the FV is employed to guide the diffusive predictor in generating human-like predictions. Therefore, we should re-train the reward function to distinguish true rewards after adding noises. After adding noises as Eq. 10 shows, we train a reward guide via the following loss function:

$$\hat{r}_t = R_\zeta(s_{k,t}) \tag{23}$$

$$L_r = \mathbb{E}||\hat{r}_t - r_t||^2 \tag{24}$$

where $s_{k,t}$ is the state at time step t with step k noise; $R_\zeta(\cdot)$ is the reward guide function parameterized by ζ; r_t is the true reward value at time step t.

6) Training With Realistic Constraint: To further ensure that the generated trajectories exhibit human-like characteristics, we introduce a realistic constraint aimed at recovering the original trajectory at each diffusion step for both diffusion models. Take the diffusive predictor as an example, the trajectory can be recovered by:

$$\hat{\tau}_0 = \frac{1}{\sqrt{\bar{\alpha}_k}} \tau_k - \sqrt{\frac{1}{\bar{\alpha}_k} - 1} \cdot \epsilon_\xi(\tau_k, k, \mathbf{c}) \tag{25}$$

Then the reconstruction error can be obtained:

$$L_s = \mathbb{E}||\hat{\tau}_0 - \tau_0||^2 \tag{26}$$

E. Inference Process

For both diffusion models, we can generate trajectories from isotropic Gaussian noises step by step. Take the diffusive predictor as an example:

$$\tau_{k-1} = \frac{1}{\sqrt{\alpha_k}} (\tau_k - \frac{\beta_k}{\sqrt{1 - \bar{\alpha}_k}} \epsilon_\xi(\tau_k, k, \mathbf{c}) + \sqrt{\beta_k} \mathbf{z}) \tag{27}$$

where $\tau_K \sim N(0, \mathbf{I})$; $\mathbf{z}$ is a random variable in the standard Gaussian distribution; ϵ_ξ is the trained model.

During the inference phase, the potential goal state is first predicted using the goal predictor. Next, we sample M goal states around $\hat{s}_T$ (denoted as $\hat{s}_{T,j}$, $j = 1, \ldots, M$) to generate planned trajectories based on Algorithm 3. Afterwards, both planned and historical trajectories are considered as conditions of the diffusive predictor. The prediction process is guided by the reward guide, as outlined in Algorithm 4. Finally, the learned LCV reward function $r_{\mathrm{LCV}}(\cdot)$ is used to evaluate the scores of each planned and predicted pair of trajectories. The trajectory pair with the highest reward is considered as the output.

IV. Experiments

In this section, we would like to verify the effectiveness of the proposed framework. First, we introduce the dataset, implementation details and metrics used in this study. To illustrate the differing emphasis of the LCV and the FV, we utilize SHAP values to analyze the feature importance of the recovered reward functions. Subsequently, we assess the accuracy

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Algorithm 3 Diffusive Planner Sampling Process

Input: Trained diffusive planner, initial and predicted goal states $s_{0,0}$ and $\hat{s}_{0,T,j}$

while *not done* **do**
  Initial trajectory $\tau_K \sim \mathcal{N}(0, \mathbf{I})$
  for $k = K, \ldots, 1$ **do**
    $\mu, \Sigma \leftarrow \mu_\xi(\tau_k), \Sigma_\xi(\tau_k)$
    $\tau_{k-1} \leftarrow$ sample from $\mathcal{N}(\mu, \Sigma)$
    $s_{k,0}, s_{k,T} \leftarrow s_{0,0}, \hat{s}_{0,T,j}$
  end for
end while

Algorithm 4 Diffusive Predictor Sampling Process

Input: Trained diffusive predictor, gradient scale γ.

$\tau_K \leftarrow$ sample from $\mathcal{N}(0, \mathbf{I})$.
for $k = K, \ldots, 1$ **do**
  $\mu, \Sigma \leftarrow \mu_\xi(\tau_k), \Sigma_\xi(\tau_k)$
  $\tau_{k-1} \leftarrow$ sample from $\mathcal{N}\left(\mu + \gamma \Sigma \nabla_{\tau_k} \log R_\zeta, \Sigma\right)$
end for

of the diffusive predictor and evaluate the performance of the diffusive planner.

Then, we conduct an ablation study and offer a visualization of the generation process. We also implement a dynamic simulation to examine lane-changing differences across various traffic contexts. Finally, we explore customized generation with general targets.

A. Dataset

In this study, we primarily evaluate and discuss the model's performance using the NGSIM dataset [44], as it is the most widely used dataset for lane-change analysis. To demonstrate the generalization ability of Diff-LC, we also present its results on the exiD dataset [45], with additional details provided in the Appendix.

NGSIM includes I-80 and US-101 datasets with a sampling rate as 10 Hz. Each dataset covers 45 mins with mild, moderate and congested traffic conditions. To identify discretionary LC events, we use the following criteria [46]: only passenger cars are included; LC events from or to on/off-ramp lanes are deemed as mandatory LC; LC events from or to lane 1 are not considered in NGSIM and datasets since lane 1 is a high occupancy lane; the LC events are not counted if there are missing surrounding vehicles.

In detail, we identify the LC beginning and ending points based on the automatic labeling method [47]. Then we extract the trajectories 2 seconds before and after LC execution for training purposes since the 2-second interval represents the standard reaction time in previous studies [1]. For both I-80 and US-101 datasets, we collect 485 LC events, and 85 events are used for evaluation.

B. Implementation Details

For the states, we normalize positions to [0, 1], and velocities and accelerations to $[-1, 1]$. Regarding MA-AIRL model, we use the proximal policy optimization (PPO) algorithm [48] as the generator. The learning rates of the actor and critic are set to 3e-4 and 1e-3 respectively. We utilize a simple two-layer MLP as the discriminators, with both their learning rates set to 3e-4. The hidden dimension is set to 64. The range of acceleration is limited to [-5, 5] m/s^2.

For the diffusive planner, the hidden dimension is set to 64, and the dimensions of convolution kernels in the U-Net are 1, 2, 4, and 8. The diffusion step is set to 20, and the learning rate is set to 2e-4. For the diffusive predictor, the hidden dim and learning rate are set to 64 and 2e-4 respectively. The diffusion step is set to 100. We utilize 2-second trajectories before lane change execution as historical trajectories, longer trajectories can be utilized if available. Both planned and prediction horizons are set to 10 seconds since the LC maneuver normally lasts for 10 seconds [49]. Our framework can also be applied for dynamic LC planning, we will reveal it in Section IV-H.

During the sampling process, we uniformly sample 20 final speed and longitudinal offset around $\hat{s}_T$, the lateral offset is set to reach the center of the target lane, and the lateral speed is set to be zero.

C. Metrics

To evaluate the accuracy of predicted trajectories, we calculate Average Displacement Error (ADE) and Final Displacement Error (FDE) for baselines and our model in the next 10-second horizons. For planned trajectories, we measure the minADE and minFDE among the top-3 most human-like trajectories and ground truth trajectories. ADE and FDE can be computed by:

$$ADE = \frac{1}{T}\sum_{t=1}^{T}\sqrt{\left(\hat{l}_{t,x} - l_{t,x}\right)^2 + \left(\hat{l}_{t,y} - l_{t,y}\right)^2} \tag{28}$$

$$FDE = \sqrt{\left(\hat{l}_{T,x} - l_{T,x}\right)^2 + \left(\hat{l}_{T,y} - l_{T,y}\right)^2} \tag{29}$$

where $l_{t,(\cdot)}$ and $\hat{l}_{t,(\cdot)}$ are ground truth and predicted/planned positions at time t, respectively.

D. Reward Function Analysis

To provide an intuitive understanding of the recovered reward functions and analyze the distinctions between the behaviors of LCV and FV, we utilize SHAP values [50] to demonstrate the importance of states. In the SHAP value plots, a point colored red (or blue) indicates a high (or low) value of the corresponding sample. Since we recover human behaviors from demonstrations, high SHAP values actually signify the behavioral tendencies of human drivers.

As Fig. 4 shows, when the LCV executes LC maneuvers, the LCV approaches the $\mathrm{LV_{tar}}$ in the longitudinal direction and minimizes the lateral distances between vehicles in the target lane. Additionally, the LCV tends to accelerate, as large speed differences between the LCV and the $\mathrm{LV_{tar}}$, as well as between the LCV and the FV, result in high rewards. As for the FV, it primarily focuses on the speed difference between the LCV and its original leading vehicle, and the distance of lateral position between the LCV and the FV. Moreover, a significant speed difference between the LCV and the FV results in a large

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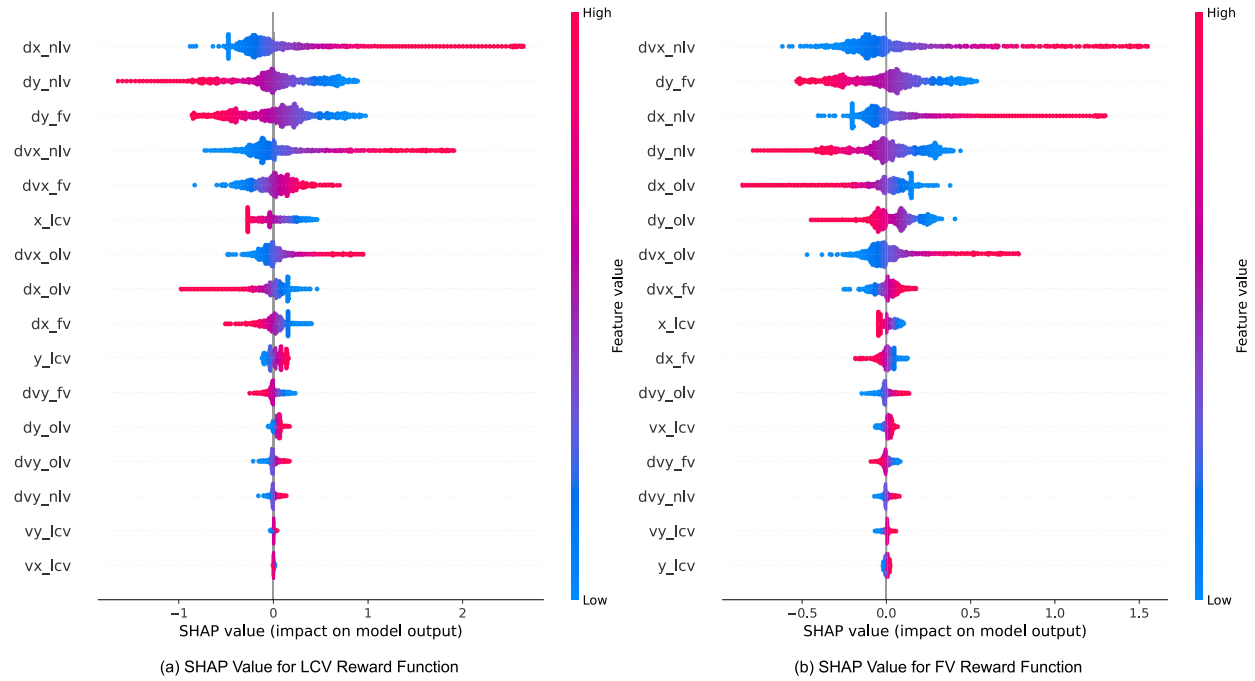

Fig. 4. SHAP values of both LCV and FV reward functions. Each point on the figure represents a single data sample within the dataset, with the color indicating the value associated with that specific data point. The horizontal axis illustrates the SHAP value, indicating the impact of the value on the model output. A positive SHAP value indicates that the corresponding data point enlarges the overall output of the model.

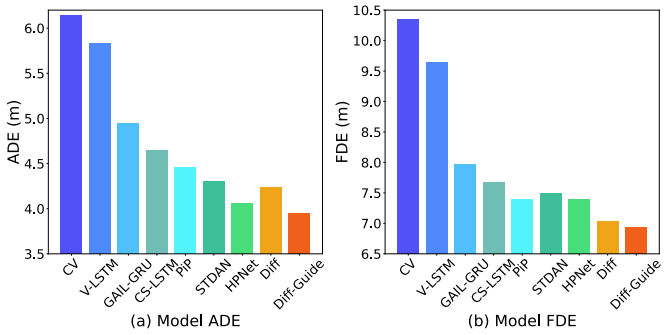

Fig. 5. Comparison of ADE and FDE for 10s trajectory prediction of the FV.

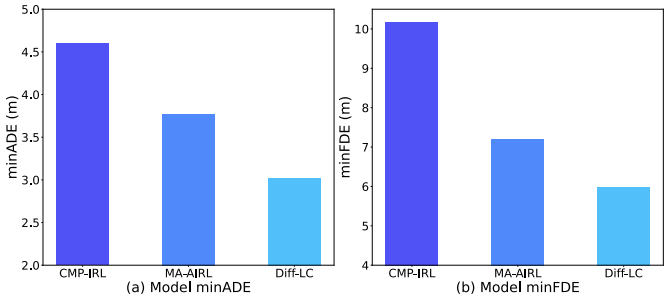

Fig. 6. Comparison of planned lane-change trajectories' human-likeness, reflected by minAED and minFDE with respects to human-drivers' behavior.

reward. Hence, the FV prefers a high velocity of the LCV and large speed differences, as it signifies high travel efficiency. For both LCV and FV, the importance of lateral speed difference is low, indicating that human drivers primarily make decisions based on distances and longitudinal speed differences. Based on the analysis of SHAP values for both reward functions, we conclude that human drivers consistently prioritize high travel efficiency. This is evidenced by the LCV tending to actively execute lane-changing maneuvers, while the FV aims to maintain high velocity.

E. Evaluation of Diffusive Predictor

Our diffusive predictor aims to predict future car-following behaviors of the FV under the impact of the LCV. To evaluate the performance, we compare it with several baselines:

- Constant Velocity (CV): CV merely uses a constant Kalman filter to predict future trajectories.
- Vanilla LSTM (V-LSTM): The future trajectories are predicted in a sequence-to-sequence manner via a vanilla LSTM.
- GAIL-GRU [51]: GAIL-GRU is based on a generative adversarial imitation learning model to predict trajectories.
- CS-LSTM [52]: CS-LSTM consists of an LSTM to handle temporal dependencies and a convolutional social pooling layer to extract spatial relations.
- PiP [53]: Similar to our model, PiP predicts the future positions of surrounding vehicles based on the planned trajectory of the AV.
- STDAN [54]: STDAN employs a multi-head attention mechanism to capture social interactions at different timestamps and utilizes a multi-head self-attention mechanism to learn the temporal dependencies between these interactions.
- HPNet [55]: HPNet designs a historical prediction attention module to capture dynamic relationship between

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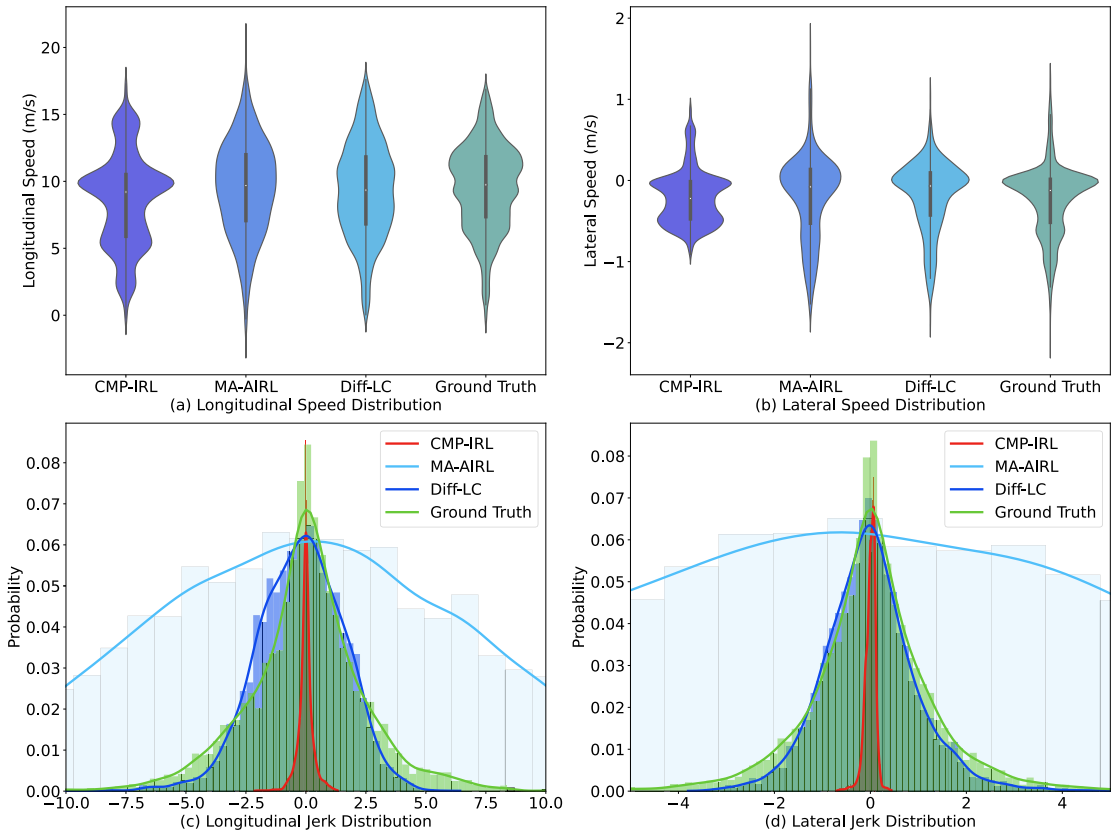
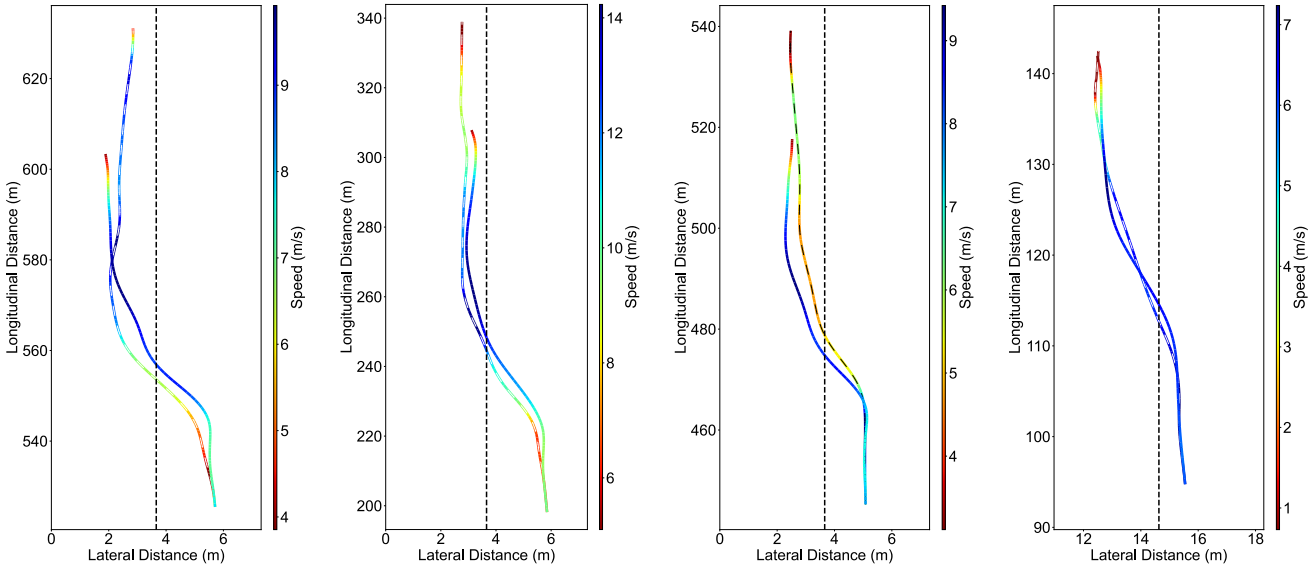


Fig. 7. Comparison of ground-truth vs. predicted speed and jerk distributions.



(a) Vehicle ID 555 (b) Vehicle ID 916 (c) Vehicle ID 943 (d) Vehicle ID 1193

Fig. 8. The generated and ground-true lane-changing trajectories. The trajectories represented with dashed lines indicate the ground truth paths, and the trajectories without dashed lines are paths planned by Diff-LC.

successive predictions, and then forecasts future trajectories based on observed data and historical predictions.

Additionally, we further list the prediction errors of the diffusive predictor with and without the reward guide. The FV diverges from typical car-following patterns as it operates under the influence of the LCV, rendering the prediction of its future behavior notably more challenging [56], [57]. Our diffusive predictor does not achieve the best performance in the ADE metric but performs well in the FDE metric. Both PiP and the diffusive predictor outperform recent models (e.g., STDAN and HPNet) in FDE, demonstrating that leveraging planned trajectories enhances the prediction of final

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displacement. Furthermore, we utilize the reward guide to control the generation process, ensuring that trajectories closely resemble human behavior. The results show that leveraging the reward guide can further enhance prediction accuracy, underscoring the superiority of the diffusion model

F. Evaluation of Diffusive Planner

In this section, we verify the effectiveness of the Diff-LC framework. For comparison, we choose a state-of-the-art framework named CMP-IRL [32] for comparison. The CMP-IRL employed a conditional motion prediction network for prediction, with planned behaviors generated through polynomial curves. Additionally, the authors employed the maximum entropy principle to evaluate trajectories based on hand-crafted features. Besides, we also use the learned actor via MA-AIRL as a baseline method for comparison. For evaluation, we quantify the minADE and minFDE between the planned trajectories and the actual trajectories. Furthermore, to demonstrate human likeness, we provide distributions of speed and jerk.

Fig. 6 presents the minADE and minFDE results. Our framework demonstrates superior performance compared to CMP-IRL, attributable to several key factors. First, we utilize the advanced diffusion model within our prediction module, coupled with precise control over the generation process through the utilization of a learned reward guide. Second, we employ MA-AIRL instead of maximum entropy IRL for both evaluation and guidance purposes, eliminating the need for hand-crafted features and enhancing expressiveness. Finally, we integrate a diffusive planner for behavior generation, enabling our model to produce behaviors that closely mimic human actions. Furthermore, the comparison between Diff-LC and MA-AIRL reveals that while IRL has been extensively employed in human-like modeling, the learned agent may not consistently generate high-quality behaviors. We also observe a collision rate of approximately 2% for our methods, determined by checking if the minimum distance between the LCV and FV/LV is less than 1 meter. This suggests that Diff-LC can generally produce realistic and safe lane-changing trajectories, though some risks remain. For inference time, the paralleled planning process of Diff-LC on NVIDIA A800 takes around 1,200 ms compared to CMP-IRL with 10 ms. Since slow inference is a primary limitation of current diffusion models [58], numerous studies have proposed methods to accelerate generation. For instance, switching to Deep Diffusion Implicit Model (DDIM) [59] could reduce computation time to around 700 ms.

Fig. 7 demonstrates the distributions of speed and jerk for three methods and the ground truth. For the speed distributions, the difference between CMP-IRL and the ground truth is the largest, while both MA-AIRL and Diff-LC can capture the rough distributions of human drivers. However, when we examine the distributions of jerks as depicted in Fig. 7 (c) and (d), it becomes evident that although MA-ARIL replicates human behavior in terms of speed, it produces excessively large and unrealistic jerks. This phenomenon can also be observed in other RL-based methods [60]. Among various methods, Diff-LC exhibits the most similar jerk distribution

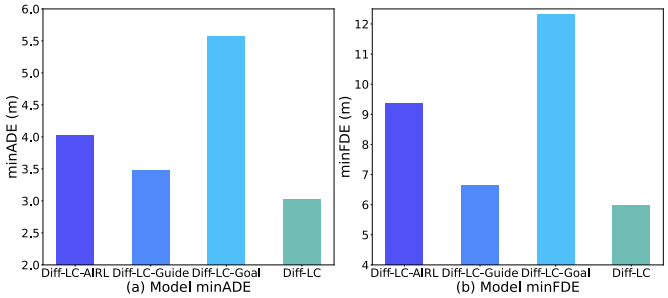

Fig. 9. Ablation study of Diff-LC framework.

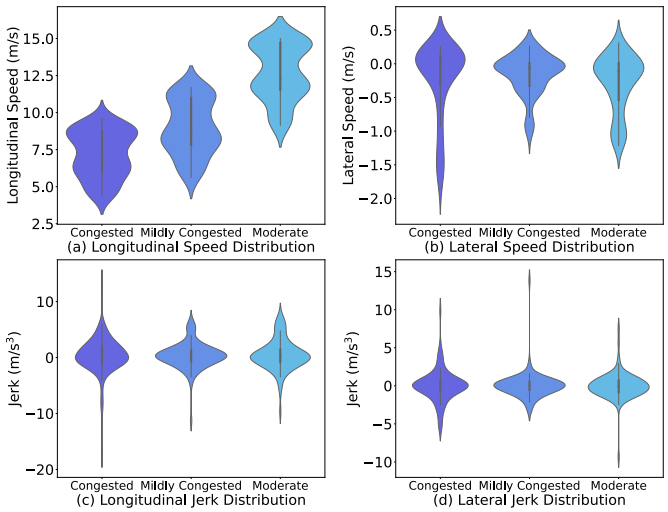

Fig. 10. Comparison of speed and jerk distributions in congested, mildly congested, and moderate conditions.

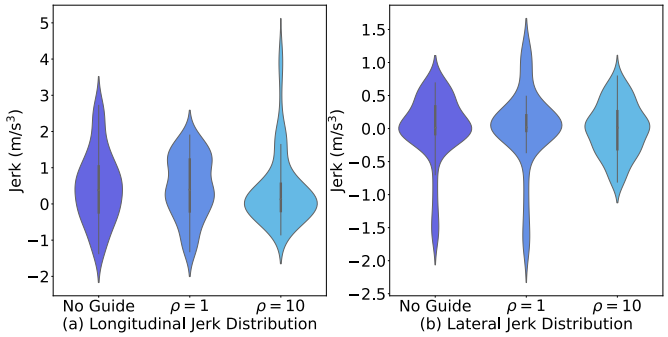

Fig. 11. Jerk distribution for varying guide strengths. The average longitudinal jerk is significantly reduced with the application of a strong guide.

to the ground truth, demonstrating that our framework can faithfully model human drivers.

Fig. 8 shows several examples of the planned trajectories, and the trajectories with dash lines indicate the ground truth. As evident, the trajectories planned by Diff-LC exhibit satisfactory similarity to the ground truth. Furthermore, it is notable that trajectories driven by human drivers are irregular and cannot easily represented by polynomial curves.

G. Ablation Study

To present the rationality of each module in our framework, we implement the ablation study in this section. Specifically,

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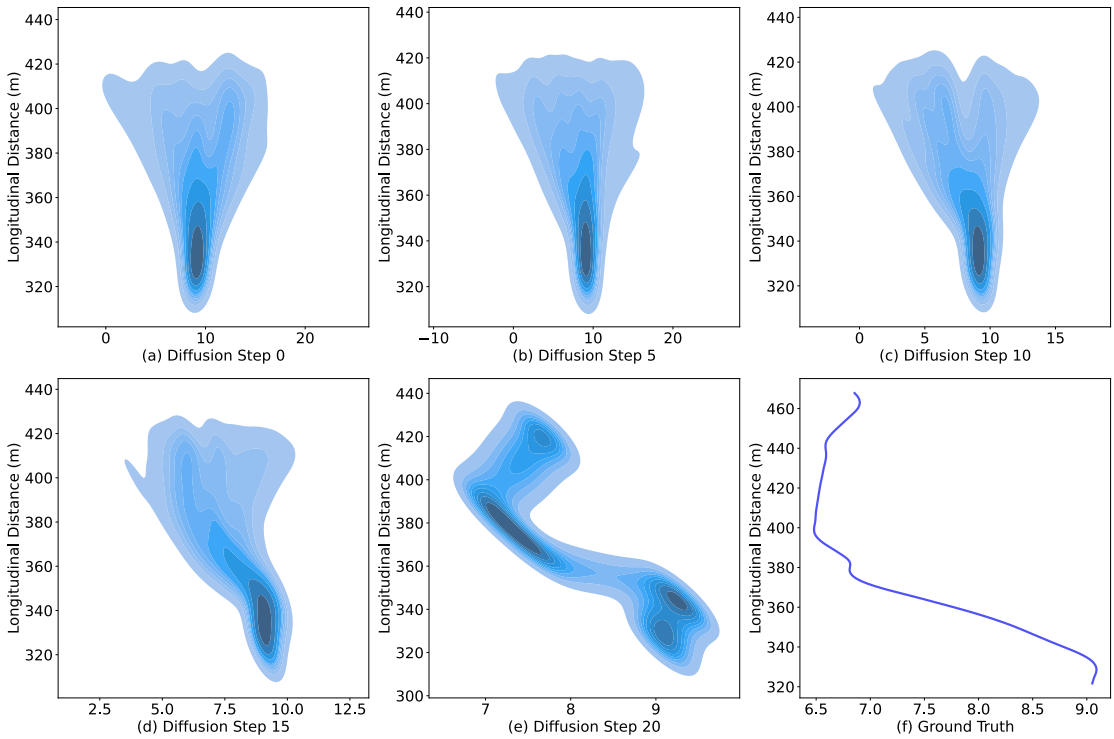

Fig. 12. Generation process of the diffusive planner.

we (i) replace MA-AIRL with AIRL, i.e. eliminate interaction effects in the reward function, denoted as Diff-LC-AIRL; (ii) eliminate reward guide in the prediction stage, denoted as Diff-LC-Guide; (iii) eliminate the goal-state predictor and uniformly sample goals from a pre-determined range, denoted as Diff-LC-Goal.

The minADE and minFDE of all ablated models are presented in Fig. 9. We observe that the removal of the guide has minimal impact on the model's performance, whereas the planner demonstrates high sensitivity to the specified goal states. This observation inspires us to focus on enhancing either the accuracy of goal prediction or the robustness of the diffusive planner. The decrease in performance resulting from replacing MA-AIRL with AIRL highlights the importance of the implicit interactions learned by MA-AIRL in achieving human-like modeling.

H. Dynamic Simulation with Various Scenarios

The analysis and results discussed above pertain to long-term planning and prediction (i.e. 10 seconds). Given the dynamic nature of real traffic environments, we expect the planner to adapt to evolving contexts. In this section, we implement a simulation with a real-time planning strategy. Specifically, we train a dynamic Diff-LC model that incrementally plans lane-change motions in 0.5-second intervals. Additionally, we use the Intelligent Driver Model (IDM) to simulate the behavior of FVs. To assess Diff-LC's sensitivity to various interaction scenarios, we compare its performance under congested, mildly congested, and moderate traffic conditions since lane-changing typically occurs across a range of traffic densities, i.e. setting the speed of two LVs to be 6 m/s, 8 m/s and 12 m/s respectively.

As shown in Fig. 10, the longitudinal speed is highest in moderate condition, followed by mildly congested and then congested conditions. Interestingly, we find that the lateral speed during lane-changing is higher in congested conditions compared to other scenarios. The primary reason is that gaps are shorter in congested traffic, prompting the LCV to move quickly from the current lane to the target lane. Additionally, all three scenarios exhibit similar levels of jerk, though some high jerk values occur due to the dynamic nature of the simulation.

I. An Attempt of Customized Generation

A key advantage of diffusion models is the ease of incorporating customized requirements during generation without re-training the model [61], [62], such as adhering to specific paths and reaching designated goal states. Recent studies in autonomous driving have proposed various approaches to enhance vehicle control and simulation, focusing on efficiency, sustainability, and safety [63]. To keep pace with the trend, we also present an attempt for customized trajectory planning.

In detail, we take the comfort factor as an example and use Diffusion Posterior Sampling (DPS) method [61], i.e. minimizing jerk during lane-changing. Based on the forward process of DDPM as Eq. 10 shows, we can recover the predicted trajectory $\hat{\tau}_0$ with Eq. 25. Afterwards, we can modify the τ_k based on the gradient of jerk:

$$\tau_k' = \tau_k - \rho \cdot \nabla \mathbf{J} \tag{30}$$

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where ρ is a hyperparameter determining the strength of guidance; $\mathbf{J}$ indicates the jerk of the predicted trajectory $\hat{\tau}_0$.

In the experiment, we set the hyperparameter to 0, 1, and 10, respectively, and calculate the jerk of the planned trajectories. Fig. 11 shows the jerk distributions for different guide strengths. The guide effectively reduces longitudinal jerks, with the average longitudinal jerk decreasing from 0.86 to 0.61 m/s^3, while lateral jerks remain around 0.35 m/s^3. Please note that our framework is designed to achieve human-like planning; therefore, the guidance can only fine-tune trajectories within human-like distributions. Besides, as mentioned in [62], current posterior sampling methods remain unstable, warranting further study.

J. Visualization of Generation Process

In this section, we present the generation process of the diffusive planner from step 0 to step 20 to provide an intuitive understanding of the diffusion model. As Fig. 12 shows, at the beginning of the generation process, the trajectory is randomly sampled from a standard Gaussian distribution, encompassing both left and right lane change directions. Nevertheless, we observe that the overall travel direction remains accurate, as we specify the initial and goal states. As the diffusion step increases, the planned trajectories progressively approximate the ground truth.

V. Discussion and Conclusion

In this study, we propose a novel framework, Diff-LC, to achieve human-like modeling of LC maneuvers. Our framework employs MA-AIRL to recover the reward functions of the LCV and the FV. We have developed a diffusive predictor and planner for trajectory prediction and planning. During the testing phase, the reward function of the FV is applied as a guide to control the prediction process, whereas the reward function of the LCV is utilized for evaluation.

We verify our framework based on NGSIM dataset. Analysis of the reward functions reveals that human drivers consistently prioritize high travel efficiency during driving. The diffusive predictor makes accurate predictions under the guidance of the reward guide. Moreover, the evaluation of planning results demonstrates that Diff-LC can not only produce human-like behaviors but also achieve highly similar speed and jerk distributions to those exhibited by human drivers. We also conduct an ablation study to demonstrate the effectiveness of each component, as well as implement dynamic simulation and customized generation. Finally, we visualize the generation process of the diffusion planner to provide an intuitive understanding of our model.

As Diff-LC demonstrates a low collision rate in our experiments, its learning-based nature prevents us from fully guaranteeing the framework's feasibility and safety. Therefore, incorporating a hard-constrained approach, such as SafeDiffuser [64], is a promising direction for future work. Another limitation of Diff-LC is that diffusion model inference requires multiple sampling steps, which may be acceptable in traffic simulation but poses a challenge for real-time decision-making applications [58]. Therefore, a more efficient, fast-response model is desirable for future development. Given its strengths and limitations, the most intuitive application of Diff-LC is in traffic simulation, where it can effectively model mixed traffic flows and assess AV performance. A reliable, human-like autonomous agent also holds promise if the aforementioned limitations can be mitigated.

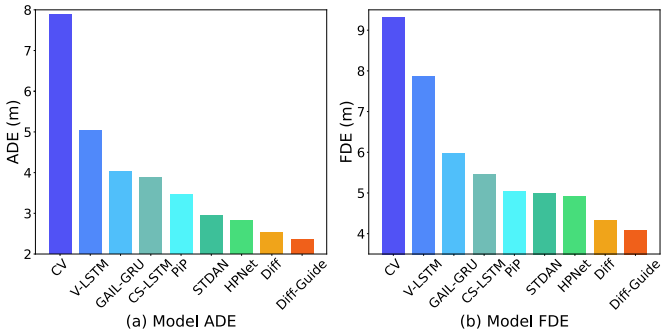

Fig. 13. Prediction results of ADE and FDE on the exiD dataset.

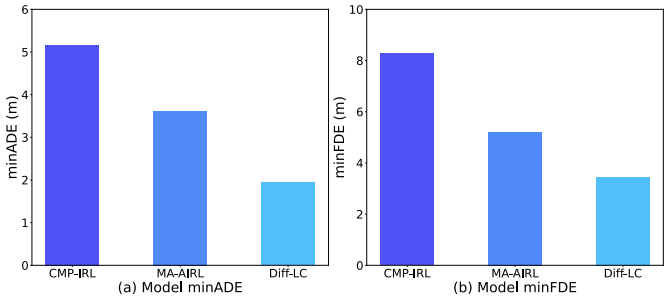

Fig. 14. Planning results of ADE and FDE on the exiD dataset.

Appendix

In the appendix, we provide the results of Diff-LC on the exiD dataset [45]. The exiD dataset offers high-precision road user trajectories at German highway entries and exits, captured via drones to overcome occlusions in traditional traffic data. Compared to the HighD dataset [65], exiD includes more LC events. Each track in the exiD dataset covers a 420-meter road segment with approximately 3,000 road users, sampled at 25 Hz. Given the relatively high driving speeds in Germany, we use the past 1-second trajectories to predict and plan motions for the next 5 seconds.

Figs. 13 and 14 compare the performance of the diffusive predictor and planner with other baseline methods. The results are consistent with the HighD dataset, with Diff-LC outperforming other baselines in both prediction and planning tasks.

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